

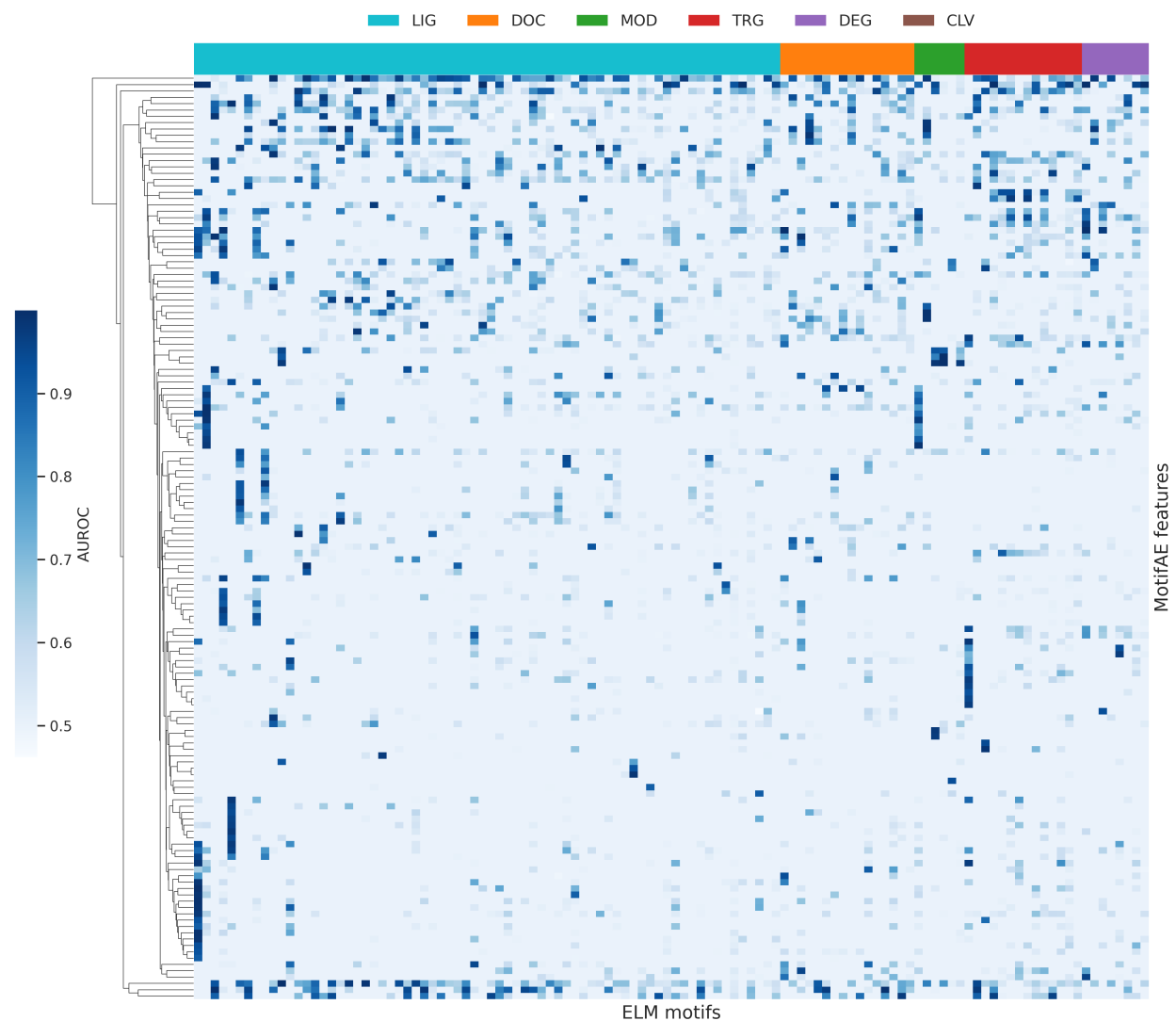


Figure S2. MotifAE feature performance on ELM motifs.

Matched motif–feature pairs were defined as those with AUROC > 0.9, resulting in 322 matched pairs involving 114 ELM motifs and 146 MotifAE features. The heatmap shows the performance (AUROC, indicated by blue intensity) of these 146 features across all 114 motifs. The six motif categories on the x-axis are the same as those shown in Figure 3A.

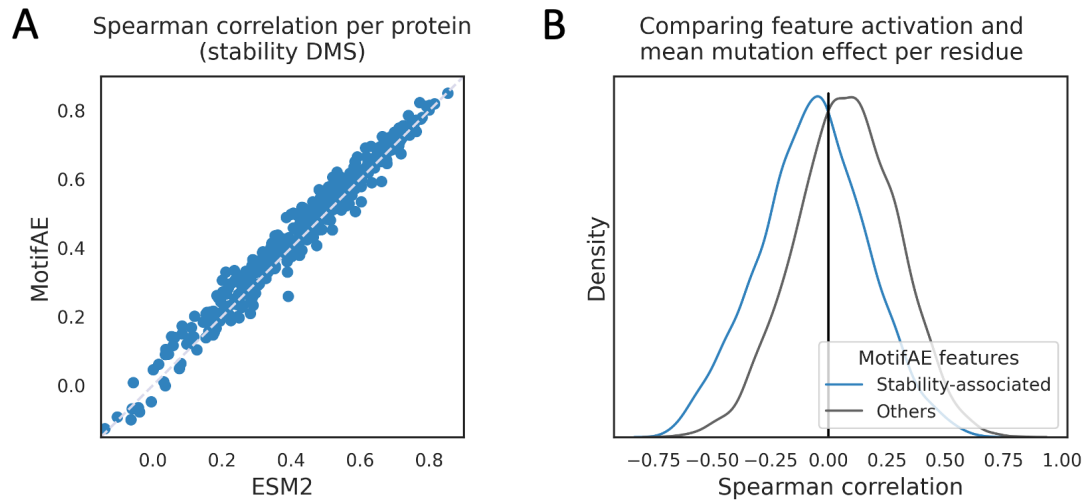


Figure S3.

(A) Zero-shot performance (Spearman correlation) on 412 protein stability DMS datasets. **(B)** Distribution of Spearman correlations between feature activation values and mean mutation effects per residue for each protein. Only features with at least ten activated residues in a protein were considered. Because destabilizing mutations have negative $\Delta\Delta G$ values, a negative Spearman correlation indicates that the feature is associated with protein stability.